Shivendra Deshpande

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.93
Ambubai Patil Junior College	Class XII (Maharashtra State Board)	2020	75.38%
Lakshminarayan Malu highschool	Class X (Maharashtra State Board)	2018	97.60%

EXPERIENCE

Research Assistant, IIT Hyderabad

- Researching high-performance **GPU-accelerated systems** for **dynamic** and memory-efficient **Approximate Nearest Neighbor Search (ANNS)** at large scale.
- Implemented **FreshDiskANN-GPU** from scratch and developed CUDA kernels for GrAND-Vamana, enabling batched insertion, in-place deletion, parallel graph pruning, and dynamic graph updates.
- Developed LowMemBANG, an SSD-resident ANNS system that minimizes DRAM footprint while enabling scalable vector search over large datasets.
- Built benchmarking infrastructure for **1M-100M vector workloads**, measuring query latency, update throughput, recall, and memory footprint across ANNS systems.
- Investigated graph-based ANNS systems, including DiskANN, Vamana, BANG, FreshDiskANN, and SVFusion.

Python Developer Intern, Sahu Technologies

- Contributed to a **Python-based web development project** using **Flask and Django**, implementing and testing application functionality.
- Gained hands-on experience in **Python development, debugging, testing, and web application maintenance**.

PROJECTS

Low-Latency Concurrent Transaction System - GitHub

- Built a multithreaded **lock-free** transaction engine using atomic synchronisation and mutexes for **concurrent execution**.
- Implemented priority-aware scheduling and contention-control mechanisms for concurrent transaction workloads.
- Achieved **12.5K operations/sec** while maintaining data consistency under concurrent execution.

GPU-Accelerated Embedding Inference & Semantic Retrieval Engine - GitHub

- Built a high-throughput GPU retrieval engine with batched transformer inference with **ONNX Runtime** and a **optimized** HNSW index.
- Achieved **8× higher throughput** than a CPU baseline at **<5 ms P99 latency** on a 1M-embedding corpus.
- Implemented dynamic batching, CUDA stream pipelining, and **FP16 quantization**, sustaining **>5K QPS** on a single NVIDIA A100.

Mixed-Precision GPU Kernel Optimization (WARP specialization) - GitHub

- Designed **warp-level resource scheduling** to overlap **FP32** and **FP64** execution across Ampere SM partitions.
- Optimized GEMM, 2MM, and 3MM PolyBench kernels, achieving **up to 2.86× speedup** on 4096×4096 matrix workloads.
- Profiled kernel execution using **NVIDIA Nsight Compute**, analyzing warp occupancy, instruction mix, and pipeline utilization.

Inter-Procedural Constant Propagation using LLVM - GitHub

- Developed a flow-sensitive inter-procedural constant propagation and transformation pass in **LLVM-14**.
- Implemented cross-function constant propagation through intraprocedural dataflow analysis.

- Reduced up to **35% of instructions** through constant substitution and elimination of redundant arithmetic operations and assignments.

PUBLICATIONS

GrAND: GPU-based Dynamic Graph Indexes for Approximate Nearest Neighbour Search

Karthik Venkatasubba, ***Shivendra Deshpande****, Shivram S., Jyothi Vedurada*

Submitted to **Proceedings of the VLDB Endowment (VLDB), 2027** (A* Conference) [Paper]

- Developed a GPU-native framework for dynamic ANNS supporting high-throughput streaming insertions and deletions.
- Designed batched graph repair and lock-free adjacency updates for dynamic graph maintenance.
- Evaluated across 7 real-world datasets and 5 streaming workloads, achieving up to **39.8× higher insertion throughput** and **25.4× higher overall throughput** compared with evaluated baselines.

RELEVANT COURSES

- **Masters:** Foundations of Machine Learning, Parallel & Concurrent Programming, GPU Programming, Advanced DSA, Program Analysis and Compiler Optimizations , Linear Optimization. **Bachelors:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Deep Learning, Natural Langauge Processing, Object Oriented Programming

SKILLS

- **Languages:** C++, C, Python, SQL
- **Systems:** Multithreading, Concurrency, Lock-Free Programming, Linux, LLVM
- **GPU/HPC:** CUDA, Parallel Computing, GPU Optimization, Nsight Compute
- **ML Systems:** PyTorch, TensorRT, ONNX Runtime, ANN Search, HNSW
- **AI/Tools:** LangChain, LangGraph, FAISS, Sentence Transformers, Git, Shell, FastAPI

ACHIEVEMENTS

- **Reliance Foundation PG Scholar**: Selected among the **Top 100** postgraduate students nationally across all disciplines
- **Qualcomm Innovation Fellowship**: India Finalist
- **KLA ×IIT-H Hackathon** 2026: Finalist, **2nd on Private Leaderboard**
- **Smart India Hackathon (SIH) 2026:** Shortlisted among the **Top 30 teams out of 200+ submissions** to represent **IIT Hyderabad** at the national-level hackathon.
- **Competitive Programming:** Global Rank **534** in LeetCode Weekly Contest 528, **LeetCode Knight (Rating: 1857)**, solved **1000+ DSA problems** across GeeksForGeeks and LeetCode.

POSITIONS OF RESPONSIBILITY

Placement Coordinator, OCS, IIT Hyderabad.

- Coordinating **placement operations for 1,000+ students**, including recruitment drives, corporate communication, and hiring logistics.

Teaching Assistant, IIT Hyderabad

- **CS5863: Program Analysis and Compiler Optimization** - Jan-May 26
- **CS3320: Compilers I** - Apr-May 26
- **CS5013: GPU Programming** - Jul-Dec 26

General Secretary, ACES, PVGCOETM

- Led a **50-member student team**, overseeing technical events, workshops, student initiatives, and organizational

activities.